

Scientific Programming Lab Viva Questions - Answers

1. What is a programming language?

A programming language is a formal language used to write instructions for a computer to perform specific tasks.

2. What are the key features of Python?

Easy syntax, interpreted, dynamically typed, object-oriented, extensive libraries, and cross-platform compatibility.

3. How does MATLAB differ from Python?

MATLAB is primarily used for numerical computing and simulations, while Python is a general-purpose language with broader applications including data science and AI.

4. What is the difference between an interpreter and a compiler?

An interpreter executes code line-by-line, while a compiler translates the entire code into machine language before execution.

5. What is a flowchart and why is it used?

A flowchart is a diagram representing a process. It is used to visualize the logic of a program.

6. What are variables and how are they declared in Python?

Variables store data. In Python, they are declared by assigning a value, e.g., `x = 10`.

7. What is the use of the `input()` and `print()` functions?

`input()` takes user input, and `print()` displays output.

8. What is a list in Python and how is it different from an array?

A list is a dynamic, heterogeneous collection. Arrays are more memory efficient and homogeneous.

9. What is the difference between a tuple and a list?

Tuples are immutable, while lists are mutable.

10. How do you create and access a dictionary in Python?

Using curly braces: `d = {'a': 1}`. Access with `d['a']`.

11. How are arrays initialized in MATLAB?

Using square brackets: `A = [1, 2, 3]`.

12. What is a 2D array? How do you traverse it using loops?

A 2D array has rows and columns. Use nested loops to traverse.

13. What are slicing operations in Python?

Extracting parts of sequences: `a[1:3]` gives elements from index 1 to 2.

14. How do you perform matrix operations in MATLAB?

Use operators like `*`, `+`, `inv()`, etc., e.g., `C = A * B`.

15. What is a root-finding method? Name two such methods.

It finds roots of equations. Examples: Bisection and Newton-Raphson methods.

16. Explain the Bisection Method in brief.

It finds roots by repeatedly dividing the interval and checking the sign of the function.

17. Write the Newton-Raphson formula.

$$x_1 = x_0 - f(x_0)/f'(x_0)$$

18. What is the difference between Bisection and Newton-Raphson methods?

Bisection is slower but guaranteed; Newton-Raphson is faster but may not always converge.

19. What is the Trapezoidal Rule and where is it used?

It approximates definite integrals by trapezoids. Used in numerical integration.

20. How do you solve a system of linear equations using matrix inverse?

If $AX = B$, then $X = \text{inv}(A) * B$.

21. What are the advantages of numerical methods?

Useful for complex problems, easy to automate, and applicable when analytical methods fail.

22. What is data analysis?

It involves inspecting, cleansing, transforming, and modeling data to extract useful information.

23. What are the functions of the pandas library?

Data manipulation, analysis, handling missing data, and reading/writing data files.

24. How do you read data from a .csv file in Python?

Using `pandas.read_csv('filename.csv')`.

25. What is the purpose of matplotlib?

It is used to create static, animated, and interactive visualizations in Python.

26. What is a histogram and what does it represent?

A histogram shows the frequency distribution of data.

27. What function is used to compute mean and standard deviation in pandas?

`mean()` and `std()` functions.

28. What is the difference between a line plot and a scatter plot?

Line plot shows trends, scatter plot shows data distribution.

29. What is IoT? Give two applications.

Internet of Things connects devices to the internet. Applications: Smart homes, health monitoring.

30. Define Machine Learning.

It is a field of AI where computers learn from data without being explicitly programmed.

31. What is Big Data?

Large, complex datasets that require advanced tools to store, process, and analyze.

32. What are the advantages of using AI in scientific programming?

Improves accuracy, handles complex data, automates tasks, and enhances decision-making.

33. Name any two project planning tools.

Gantt chart, Trello.

34. What are the key sections of a scientific report?

Abstract, Introduction, Methodology, Results, Discussion, Conclusion, References.